

1. For each equation below, write the letter that represents the best name of that 3D figure. [1 pt each]

A: Line

B: Two lines

C: Ellipsoid

D: Plane

E: Two planes

F: Elliptic Cone

G: Hyperboloid of 1 Sheet

H: Hyperboloid of 2 Sheets

I: Hyperbolic Paraboloid

J: Elliptic Paraboloid

K: Elliptic Cylinder

L: None of the above

$$y^2 - x^2 = z \quad \underline{\hspace{2cm}}$$

$$\frac{x^2}{4} + y^2 - \frac{z^2}{9} = 0 \quad \underline{\hspace{2cm}}$$

$$z = 2x + 3 \quad \underline{\hspace{2cm}}$$

$$\frac{x^2}{4} + y^2 - \frac{z^2}{9} = 1 \quad \underline{\hspace{2cm}}$$

$$x^2 + z^2 = y - 1 \quad \underline{\hspace{2cm}}$$

$$\frac{x^2}{4} + y^2 - \frac{z^2}{9} = -1 \quad \underline{\hspace{2cm}}$$

2. Consider the hyperbolic paraboloid $y^2 - z^2 = -x$. Write the letter that best represents each trace below. [1 pt each]

S: parabola that opens in the direction of the positive x-axis

T: parabola that opens in the direction of the negative x-axis

U: hyperbola that opens in the direction of the y-axis

V: hyperbola that opens in the direction of the z-axis

W: two intersecting lines

i) Trace at $y = 0$ is _____

iii) Trace at $x = 0$ is _____

ii) Trace at $z = 0$ is _____

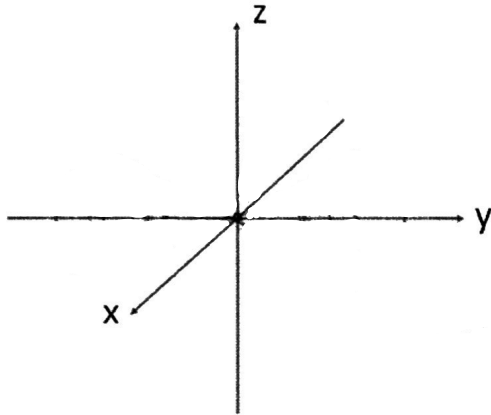
iv) Trace at $x = 1$ is _____

3. Sketch each 3D figure below. Make sure that your sketch clearly includes:

- traces in at least 2 directions should be visible
- one key point (vertex / intercept / saddle point) labeled with its coordinates
- Then, state the name of each figure.

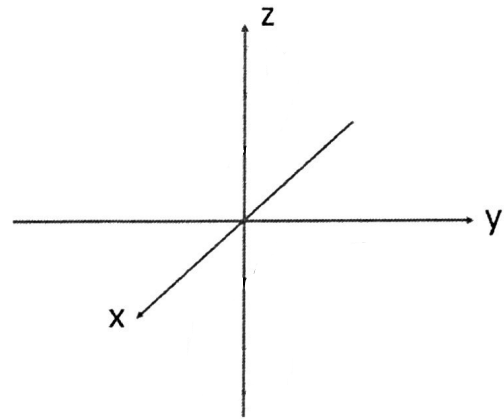
[3 pts each sketch, 1 pt name]

a) $\frac{y^2}{16} = \frac{x^2}{9} + \frac{z^2}{9}$



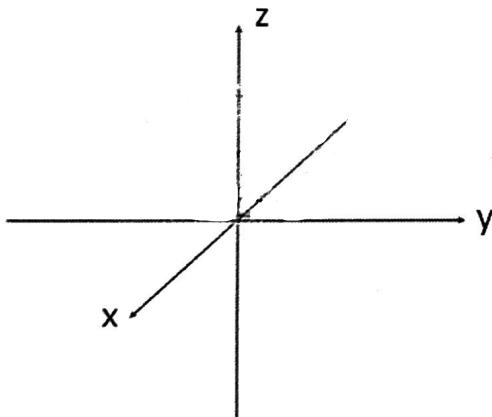
Name: _____

b) $36z^2 - 4y^2 - 9x^2 = 36$



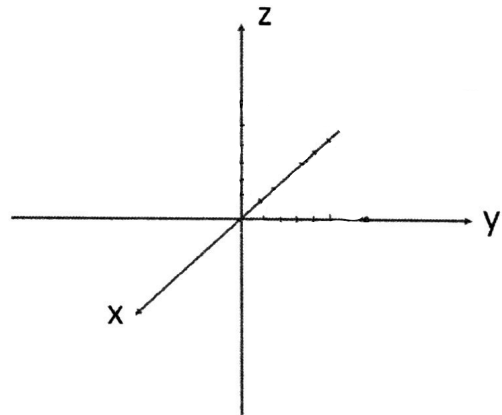
Name: _____

c) $\frac{x^2}{4} + \frac{y^2}{9} + (z - 2)^2 = 1$



Name: _____

d) $-6x + 6y + 7z = 42$



Name: _____

4. Write an equation of an elliptic paraboloid with vertex at $(0,1,0)$ that opens along the negative y-axis. [3 pts]

5. Enlighten your Analysis Honors teachers, please. What is the "six-seven" thing about? What does it mean TO YOU? [0 pts]